```
D1  <- samps1[[1]]


model2 <- jags.model(textConnection(model_string2),
                     data = list(Y=Y,N=N,n=n,X=X,Z=Z),n.chains=1,
                     quiet=TRUE)
update(model2, 10000, progress.bar="none")
samps2 <- coda.samples(model2,
          variable.names=c("D","beta"),
          n.iter=20000, progress.bar="none")
plot(samps2)
```



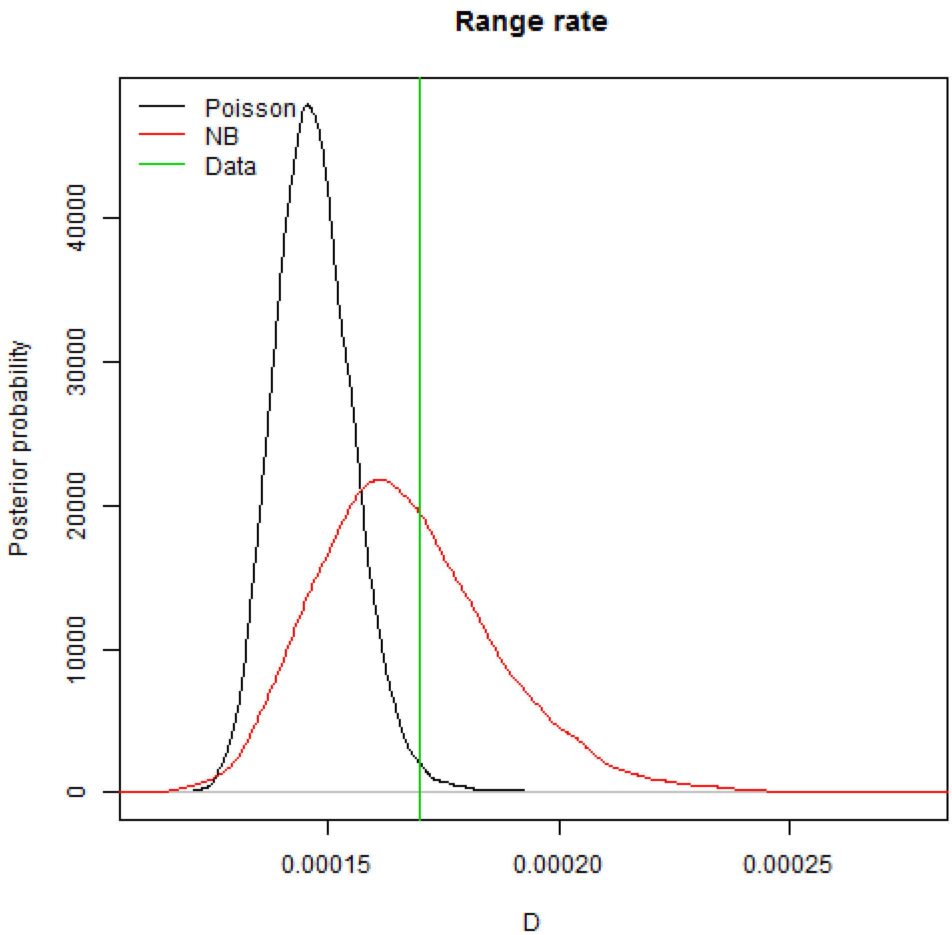
```
# Compute the test stats for the data
rate <- Y/N
D0   <- c(   min(Y),   max(Y),      max(Y)-min(Y),
          min(rate),max(rate),max(rate)-min(rate))
Dnames <- c("Min Y", "Max Y", "Range Y", "Min rate", "Max rate", "Range r

# Compute the test stats for the models

pval1 <- rep(0,6)
names(pval1)<-Dnames
pval2 <- pval1

for(j in 1:6){
  plot(density(D1[,j]),xlim=range(c(D0[j],D1[,j],D2[,j])),
                      xlab="D",ylab="Posterior probability",
                      main=Dnames[j])
 lines(density(D2[,j]),col=2)
 abline(v=D0[j],col=3)
 legend("topleft",c("Poisson","NB","Data"),lty=1,col=1:3,bty="n")

 pval1[j] <- mean(D1[,j]>D0[j])
 pval2[j] <- mean(D2[,j]>D0[j])
}
```



5. Results

```
pval1
```

```
##       Min Y      Max Y    Range Y   Min rate   Max rate Range rate
##     0.92440    0.02100    0.01610    0.98635    0.06800    0.01125
```

```
pval2
```

```
##       Min Y      Max Y    Range Y   Min rate   Max rate Range rate
##     0.79130    0.34375    0.33545    0.92460    0.56570    0.40190
```

The regular Poisson model has several p-values near zero or one, so it doesn't seem to fit well. The negative binomial model fits better.